

Probabilistic Topology Evolution and CH Optimization using CDO in WSNs

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Introduction to Wireless Sensor Networks (WSNs)

- **WSNs consist of small, autonomous sensor nodes** that monitor environmental or physical conditions like temperature, pressure, or motion.
- **Sensor nodes communicate wirelessly** and forward collected data to a central base station for processing.
- **They are used in various applications** such as smart agriculture, environmental monitoring, healthcare, and military systems.
- **Nodes are typically battery-powered**, making energy efficiency a top design priority.
- **Efficient communication and clustering protocols** are vital to ensure scalability, reliability, and extended network lifetime.

Objectives

- To develop an energy-efficient communication model for Wireless Sensor Networks (WSNs).
- To optimize Cluster Head (CH) selection using Cognitive Decision Optimization (CDO), based on energy, connectivity, and distance.
- To implement Probabilistic Topology Evolution (QTBE) for dynamic and adaptive link formation.
- To ensure balanced energy consumption across the network to delay node death and extend network lifetime.
- To build a scalable and fault-tolerant system that adapts to energy and topological changes in each round.

Importance of efficient clustering and topology management

Reduces Energy Consumption: Organizing nodes into clusters minimizes redundant transmissions and lowers communication energy costs.

Extends Network Lifetime: Balanced energy usage through optimized cluster head selection helps prevent early node failures.

Improves Scalability: Hierarchical clustering allows the network to efficiently handle a large number of nodes.

Enhances Reliability and Coverage: Proper topology ensures all areas are monitored and maintains connectivity even during node failures.

Cluster Formation and Topology Evolution

- **Cluster formation groups nearby nodes under a Cluster Head (CH)** to reduce communication distance and conserve energy.
- **CHs aggregate data from their member nodes** and forward it to the base station, minimizing redundant transmissions.
- **Topology evolution involves dynamically adjusting node connections** based on factors like distance, energy, and network changes.
- **Efficient clustering and adaptive topology** improve load balancing, fault tolerance, and overall network longevity.

What are the problems with already existing methods and why we need a new one

Random CH selection in traditional methods (e.g., LEACH) leads to unbalanced energy usage and early node deaths.

Fixed or static topologies lack adaptability, making them inefficient in dynamic environments or with node failures.

Most methods ignore key trade-offs like the combined impact of energy, connectivity, and communication distance.

Data routing paths are often suboptimal, causing unnecessary energy drain and reduced network lifetime.

A new model is needed to intelligently select CHs, evolve topology probabilistically, and route data efficiently using current network conditions.

Probabilistic Topology Evolution (QTBE)

Inspired by quantum tunneling behavior

We look the connections between the nodes as tunnels and quantum ha the quality that ensures that we can have multiple paths at the same time we kept tunneling factor to decide which connections need to be done

Tunneling factor (T): $T(i, j) = \exp(-d(i, j)/R)$

$d(i, j)$ represents distance between two nodes

The parameter R controls the rate at which the connection probability decays with distance, effectively determining the network's connectivity density.

The exponential function is used because it naturally models rapid decay in connection probability with distance, it will have a smooth transmission means higher difference between closer nodes and farther nodes

Cluster Head Optimization Using CDO

Cognitive Decision Optimization (CDO) is a method for selecting Cluster Heads based on a utility function that combines residual energy, connectivity, and average distance to ensure energy-efficient and balanced clustering.

- CHs selected based on:
 - **Energy levels**
 - **Connectivity degree**
 - **Average distance to other CHs**
- Utility function:
$$Utility = \alpha * Energy + \beta * Connectivity - \gamma * AvgDistance$$
- Weights (α , β , γ) are based on how important the parameters are

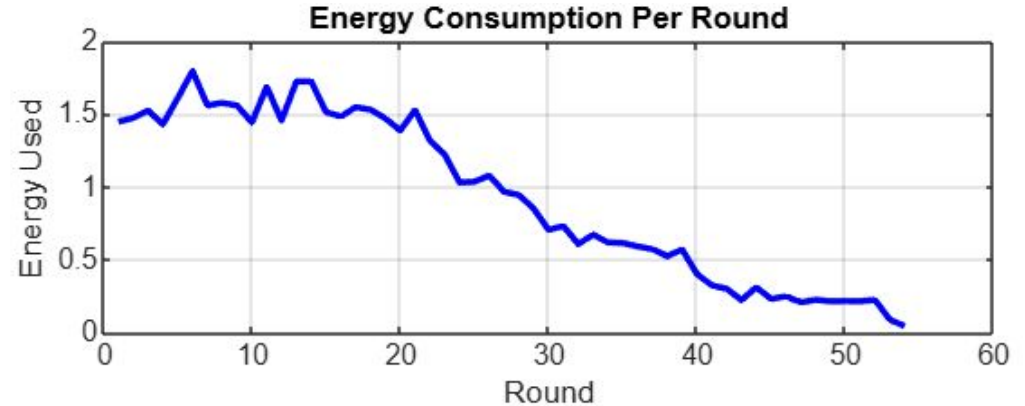
Minimum Spanning Tree (MST) for Routing

- **MST connects all Cluster Heads (CHs) and the Base Station** with the minimum total communication distance.
- **It avoids redundant and long links**, reducing energy consumption during inter-cluster communication.
- **Kruskal's algorithm is commonly used** to construct the MST efficiently.
- **MST ensures loop-free and optimal routing paths**, balancing the load among CHs.
- **Using MST extends network lifetime** by minimizing the total energy used for data transmission.
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Energy Depletion Model

Content:

- **Initial energy:** Same for all nodes
- Energy consumed during:
 - Sensor to CH transmission
 - CH to BS transmission
- Nodes die when energy $\leq$ threshold
- Simulation runs until **last node dies**

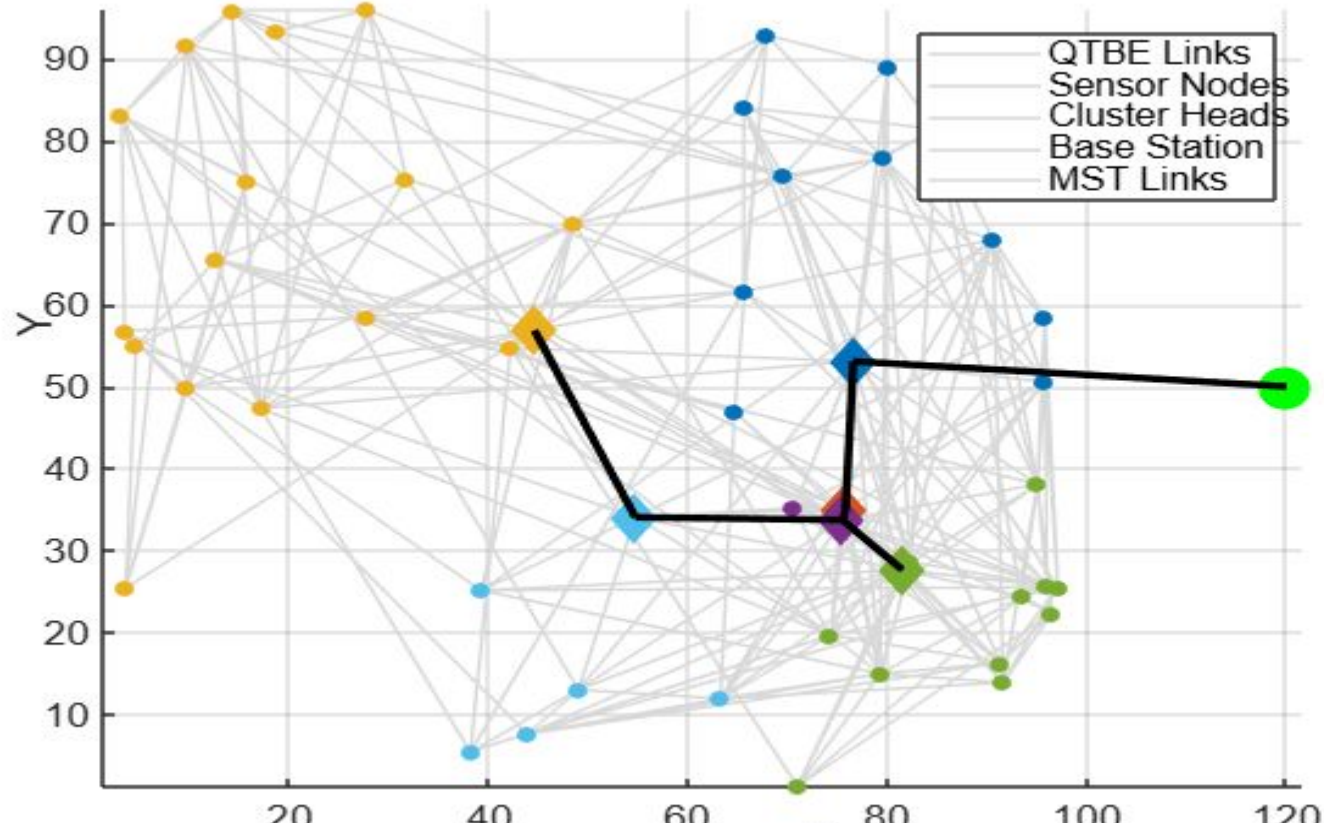


Simulation Setup

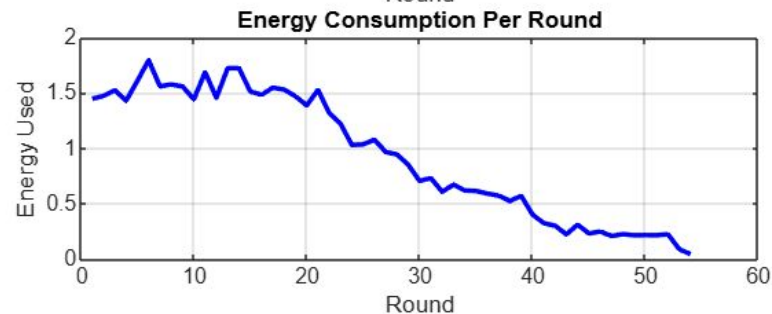
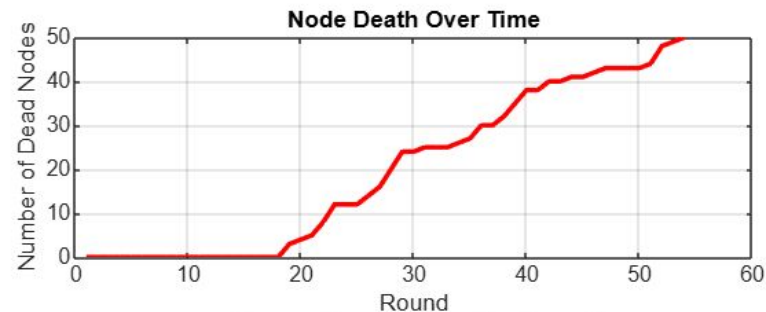
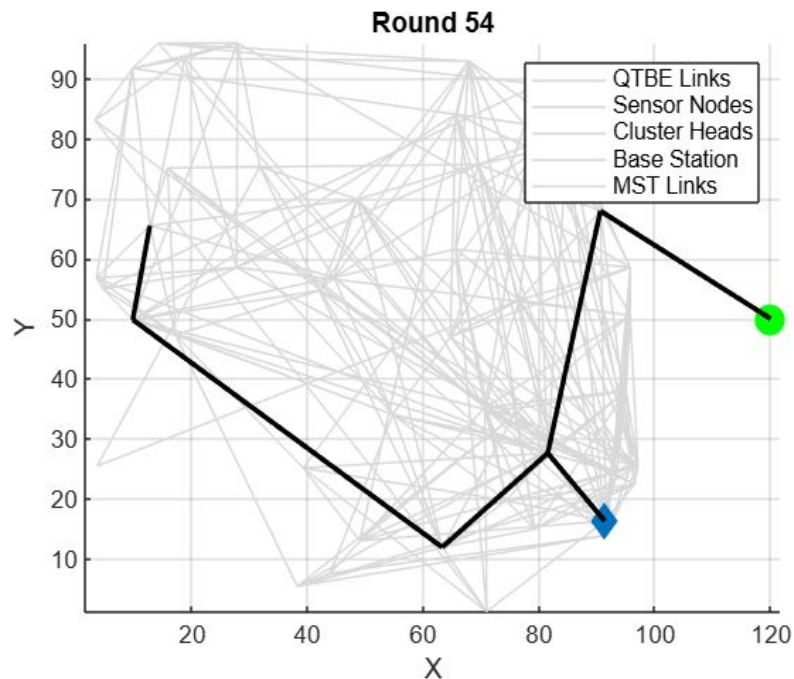
Content:

- **Number of Nodes:** 50
- **Field Size:** 100x100
- **BS Position:** Outside field (120, 50)
- **Rounds:** Until all nodes die
- **Tools:** MATLAB simulation

Topology evolved and Cluster Heads



Results and Analysis



Conclusion and Future Work

- **Conclusion:**

- Probabilistic topology provides better adaptability
- CDO improves CH selection balance
- MST enhances communication efficiency

- **Future Work:**

- Real-time adaptability to enhance performance in dynamic WSN environments.
- Explore hybrid energy harvesting models